

秦川

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主页信息

- 个人主页: <https://chuanqin8.github.io/>
- Google Scholar: <https://scholar.google.com/citations?user=17NMHvUAAAAJ>
- GitHub: <https://github.com/chuanqin8>

教育背景

兰州大学 博士 (硕博连读) · 计算机应用技术

2021.09 – 2027.06 (预期)

哈尔滨工业大学 本科 · 船舶与海洋工程

2017.09 – 2021.06

科研项目

高效锐度感知优化: SAM-LA / SAM-L

IEEE/CAA JAS 录用待刊 · NeurIPS 2026 在审

- 针对 SAM 优化器约 2 倍训练开销的问题, 提出 $O(1)$ 计算复杂度的平坦度度量——损失加速度 (LA) 与损失差 (Loss Gap), 并在梯度流框架下证明其可直接刻画优化轨迹上的损失景观锐度, 替代计算开销不可承受的 Hessian 类度量。
- 据此构建 SAM 与 SGD 自适应切换的优化器: 仅在参数穿越尖锐区域时启用 SAM, 平坦区域回退高效 SGD; 在 CIFAR / ImageNet 上以大幅降低的训练时间与计算开销取得与 SAM 相当或更优的泛化性能, 并给出收敛性理论分析。

自适应梯度调制优化方法: GAF / AGR / FedAvg-AGA

Neural Networks 2023 · IEEE TCE 2025 · IEEE TNNLS 在审

- 提出梯度激活函数对梯度进行自适应非线性重塑: 放大微小梯度分量、约束过大梯度, 缓解梯度消失/爆炸与病态问题, 显著加速深层 LSTM 收敛; 谷歌学术引用 75 次, 代码开源。
- 揭示 SAM 梯度的统计特性 (小梯度分量更少、方差与范数更大) 及其逃离尖锐极值的机制, 提出自适应梯度重塑 (AGR) 渐进放大微小梯度分量, 可同时增益 SGD 与 SAM; Hessian 最大特征值与迹的分析证实优化器收敛至更平坦的极小值。
- 将梯度激活引入联邦学习客户端训练 (FedAvg-AGA / FedSAM-AGA), 缓解非独立同分布消费电子设备数据下的训练不稳定与过拟合问题, 提升联邦学习性能并提供收敛性分析。

微分方程诠释的神经网络结构设计

IEEE TETCI / TNNLS 在审 · PNAS 投稿中

- 将残差网络前向传播视为常微分方程的 Euler 离散, 建立 Adams–Bashforth 多步离散截断误差与网络训练损失的对应关系, 提出“数值一致性是网络拟合能力与抑制退化问题的关键”的理论判据, 并在 CIFAR10/100 与 ImageNet 上验证。
- 从采样定理视角提出“近似采样”与“网络前向采样频率”概念, 设计 Taylor k 步 (TLk) 连接替代残差连接; 实验表明满足近似采样并提高采样频率可稳定提升 ResNet 与 ViT 性能、平坦化损失景观。
- 进一步提出动力学系统–信息论–深度学习的“三元对偶”假设, 统一三类系统中的全局/局部误差规律, 为理解深度网络黑箱提供新框架。

神经动力学优化与机器人/消费电子应用

IEEE TCYB / TFS / TCE / CAAI TIT 在审

- 提出梯度范数惩罚的协同神经动力学优化 (CNOGNP), 以低开销梯度范数作为锐度代理高效定位平坦极小值, 分类实验验证泛化能力提升。
- 提出模糊符号协同神经动力学 (FSCNO): 符号动量实现高精度局部搜索, 模糊控制器按种群多样性实时调度学习率以平衡全局探索; 在软体机械臂与冗余刚性机械臂的运动控制实物实验中验证有效。
- 提出量子启发协同神经动力学 (QICNO): 量子隧穿式梯度变换局部搜索 + 动态吸引子与随机跳跃全局搜索, 在 4 个交通流基准数据集上精度与收敛速度均优于主流优化算法, 面向车载导航等消费电子场景。
- 提出自调节抗扰神经动力学 (STPAN) 求解动态多线性四元数张量方程, 在线辨识并自适应补偿扰动, 完成冗余机械臂轨迹跟踪任务。

- 系统综述基于 LLM 的 AI Agent：解构推理、感知、记忆、规划、行动、学习六大核心模块及其交互机制，梳理从数字域到具身系统的应用谱系，指出关键挑战与研究路线图。

基金项目

- 基于神经动力学的深度神经网络泛化研究——甘肃省高校研究生“创新之星”项目，1 万元，项目负责人
2026.01–2026.12
- 微分方程诠释的大模型泛化和鲁棒研究——CCF-华为胡杨林基金理论计算机科学专项，11 万元，主要参与人（负责申请书撰写、中期考核与结项）
2024.10–2025.10
- 飞桨平台上微分方程诠释大模型的稳定性、收敛性及优化策略研究——CCF-百度松果基金，15 万元，主要参与人（负责项目全流程管理）
2023.11–2024.10
- 基于 MindSpore 的微分方程诠释的深度神经网络结构设计与优化器构建——CAAI-华为 MindSpore 学术奖励基金，9 万元，主要参与人（参与核心算法设计与实验验证）
2022.11–2023.11

荣誉奖项

- 年会论文一等奖
- 学业奖学金一等奖
- 兰州大学优秀研究生

其他

- 半程马拉松完赛经历：2024 兰州马拉松（01:42:38）、2024 营山马拉松（01:53:39）

论文

1. Chuan Qin and Long Jin, "A Survey of LLM-Based AI Agents," *Tsinghua Science and Technology*, in press, DOI: 10.26599/TST.2026.9010072.
2. Chuan Qin, Liangming Chen, Zangtai Cai, Mei Liu, and Long Jin, "Long short-term memory with activation on gradient," *Neural Networks*, vol. 164, pp. 135–145, 2023.
3. Chuan Qin, Mei Liu, and Long Jin, "Adaptive gradient activation for federated learning on consumer electronic devices," *IEEE Transactions on Consumer Electronics*, vol. 71, no. 2, pp. 5489–5500, 2025.
4. Chuan Qin, Liangming Chen, and Long Jin, "SAM-LA: Efficient sharpness-aware minimization training via loss acceleration," *IEEE/CAA Journal of Automatica Sinica*, accepted with minor revision.
5. Chuan Qin, Liangming Chen, and Long Jin, "Adaptive gradient reshaping boosting generalization," *IEEE Transactions on Neural Networks and Learning Systems*, minor revision.
6. Chuan Qin and Long Jin, "Anti-perturbation solution of dynamic multilinear quaternion tensor equations with application to manipulators," *CAAI Transactions on Intelligence Technology*, major revision.
7. Chuan Qin and Long Jin, "Collaborative neurodynamic optimization with gradient norm penalty for generalization improvement," *IEEE Transactions on Cybernetics*, reject & resubmit.
8. Chuan Qin and Long Jin, "Fuzzy sign-based collaborative neurodynamic optimization with application to robots," *IEEE Transactions on Fuzzy Systems*, under review.
9. Chuan Qin, Liangming Chen, Yongji Guan, and Long Jin, "SAM-L: Dynamically balancing sharpness-awareness and efficiency with loss gap metric," *NeurIPS 2026*, under review.
10. Chuan Qin, Jianfeng Lv, and Long Jin, "Quantum-inspired collaborative neurodynamic optimization for traffic flow prediction," *IEEE Transactions on Consumer Electronics*, under review.
11. Chuan Qin, Liangming Chen, and Long Jin, "Consistency is key of fitting and degradation alleviating for deep neural networks," *IEEE Transactions on Neural Networks and Learning Systems*, under review.
12. Chuan Qin, Mei Liu, and Long Jin, "Understanding residual neural networks from sampling theorem perspective," *IEEE Transactions on Emerging Topics in Computational Intelligence*, under review.
13. Chuan Qin, Long Jin, and Zhigang Zeng, "On ternary duality of dynamical systems, information theory, and deep learning," *PNAS*, submitted.

A Survey of LLM-Based AI Agents

Tsinghua Science and Technology / In Press

Chuan Qin, Long Jin

- **Abstract:** We provide a holistic review of LLM-based AI agents, deconstructing the core architecture—reasoning, perception, memory, planning, action, and learning—and examining the interplay among these modules. We further survey their transformative impact across applications from digital domains to embodied systems, and conclude with critical challenges and future directions as a roadmap for next-generation LLM-based agents.

Long Short-Term Memory with Activation on Gradient

Neural Networks, 2023 / Citation: 75 (Google Scholar)

Chuan Qin, Liangming Chen, Zangtai Cai, Mei Liu, Long Jin

- **Abstract:** We apply gradient activation—modifying gradients with a designed gradient activation function—to LSTM training, and provide empirical criteria for choosing its hyperparameters. The method alleviates the vanishing/exploding gradient and ill-conditioned problems and accelerates the convergence of LSTM.
- **Code:** <https://github.com/LongJin-lab/ACT-In-NLP>

Adaptive Gradient Activation for Federated Learning on Consumer Electronic Devices

IEEE Transactions on Consumer Electronics, 2025 / Citation: 10 (Google Scholar)

Chuan Qin, Mei Liu, Long Jin

- **Abstract:** We propose FedAvg-AGA and FedSAM-AGA, which introduce adaptive gradient activation into client-side federated training on consumer electronic devices. The methods alleviate vanishing/exploding gradients, help client models escape sharp minima under non-IID data, and improve federated learning performance, with convergence analysis provided.

SAM-LA: Efficient Sharpness-Aware Minimization Training via Loss Acceleration

IEEE/CAA Journal of Automatica Sinica / Accepted with Minor Revision

Chuan Qin, Liangming Chen, Long Jin

- **Abstract:** We propose loss acceleration (LA), an $O(1)$ sharpness measure that directly captures the loss landscape's sharpness along the optimization trajectory under the gradient flow setting. Building on LA, SAM-LA adaptively alternates between SAM and SGD, achieving generalization comparable to or better than SAM while substantially reducing training time and computational overhead, with convergence analysis provided.

Adaptive Gradient Reshaping Boosting Generalization

IEEE Transactions on Neural Networks and Learning Systems / Minor Revision

Chuan Qin, Liangming Chen, Long Jin

- **Abstract:** We demonstrate that SAM gradients contain fewer small elements and exhibit considerably greater variance and norm than SGD gradients, suggesting an enhanced ability to circumvent sharp minima. Inspired by this, adaptive gradient reshaping (AGR) progressively enlarges minor gradient elements, boosting the sharp-minima-escaping ability of both SGD and SAM, corroborated by Hessian trace/max-eigenvalue and loss-landscape analyses on CIFAR and ImageNet.

Anti-Perturbation Solution of Dynamic Multilinear Quaternion Tensor Equations With Application to Manipulators

CAAI Transactions on Intelligence Technology / Major Revision

Chuan Qin, Long Jin

- **Abstract:** We devise a self-tuning perturbation-aware neurodynamics (STPAN) method for solving dynamic multilinear quaternion tensor equations under perturbations. The solver identifies the temporal profile of disturbances on the fly and adaptively offsets them; convergence and disturbance-estimating capacity are formally established, with numerical case studies and a redundant-manipulator trajectory-tracking task confirming its practical merit.

Collaborative Neurodynamic Optimization with Gradient Norm Penalty for Generalization Improvement

IEEE Transactions on Cybernetics / Reject & Resubmit

Chuan Qin, Long Jin

- **Abstract:** We introduce collaborative neurodynamic optimization with a gradient-norm-penalized fitness function (CNOGNP), which efficiently finds flat minima using the low-cost gradient norm as a sharpness proxy. Theoretical analysis and simulations demonstrate effective flat-minima seeking, and classification experiments validate improved generalization over competing algorithms.

Fuzzy Sign-Based Collaborative Neurodynamic Optimization With Application to Robots

IEEE Transactions on Fuzzy Systems / Under Review

Chuan Qin, Long Jin

- **Abstract:** We propose fuzzy sign-based collaborative neurodynamic optimization (FSCNO), integrating sign-based momentum dynamics for precise local search with a collaborative global-exploration mechanism mediated by a fuzzy logic controller that adapts the learning rate to real-time population diversity and optimization progress. Evaluations on high-dimensional benchmarks and kinematic control of a soft robot and a redundant rigid robot—in both simulations and physical experiments—validate its efficacy.

SAM-L: Dynamically Balancing Sharpness-Awareness and Efficiency With Loss Gap Metric

NeurIPS 2026 / Under Review

Chuan Qin, Liangming Chen, Yongji Guan, Long Jin

- **Abstract:** We propose SAM-L, which dynamically switches between SAM and SGD based on the loss gap—a computationally inexpensive $O(1)$ flatness measure approximating the gradient norm. SAM-L significantly reduces computational overhead by cutting the number of SAM iterations while achieving comparable or even superior generalization, with convergence analysis and evaluations on CIFAR and ImageNet.

Quantum-Inspired Collaborative Neurodynamic Optimization for Traffic Flow Prediction

IEEE Transactions on Consumer Electronics / Under Review

Chuan Qin, Jianfeng Lv, Long Jin

- **Abstract:** We introduce quantum-inspired collaborative neurodynamic optimization (QICNO), combining a local search whose nonlinear gradient transformation acts analogously to quantum tunneling with a global search guided by dynamically computed attractors and stochastic jumps. On four benchmark traffic datasets with several forecasting models, QICNO consistently outperforms standard optimization algorithms in both accuracy and convergence speed.

Consistency Is Key of Fitting and Degradation Alleviating for Deep Neural Networks

IEEE Transactions on Neural Networks and Learning Systems / Under Review

Chuan Qin, Liangming Chen, Long Jin

- **Abstract:** We connect the consistency in numerical analysis with neural networks' performance: viewing a residual network as the 1-step Adams-Bashforth discretization of an ODE, we find the discretization's truncation error corresponds to the network loss, making consistency crucial to fitting ability and degradation alleviation. Experiments on CIFAR10, CIFAR100, and ImageNet confirm the finding, and show that without consistency networks are more prone to vanishing/exploding gradients and feature oscillation.
- **Code:** <https://github.com/LongJin-lab/consistency>

Understanding Residual Neural Networks From Sampling Theorem Perspective

IEEE Transactions on Emerging Topics in Computational Intelligence / Under Review

Chuan Qin, Mei Liu, Long Jin

- **Abstract:** We bridge residual connections and network performance from the perspective of the sampling theorem: when a network's discretization satisfies approximate sampling, increasing the network forward sampling frequency improves performance. We design Taylor k-step (TLk) connections extending residual connections, and experiments on TLk networks and vision Transformers show that ensuring approximate sampling guarantees trainability while increasing the frequency enhances the loss landscape's flatness, which is related to generalization.

On Ternary Duality of Dynamical Systems, Information Theory, and Deep Learning

PNAS / Submitted

Chuan Qin, Long Jin, Zhigang Zeng

- **Abstract:** We propose a ternary duality hypothesis linking dynamical systems, information theory, and deep learning under a consistent framework of using discrete states to approximate exact continuous ones. Three predictions—on global errors, optimal intervals, and local-error effects—are supported by empirical phenomena across all three systems, offering a unified lens for understanding deep neural networks.